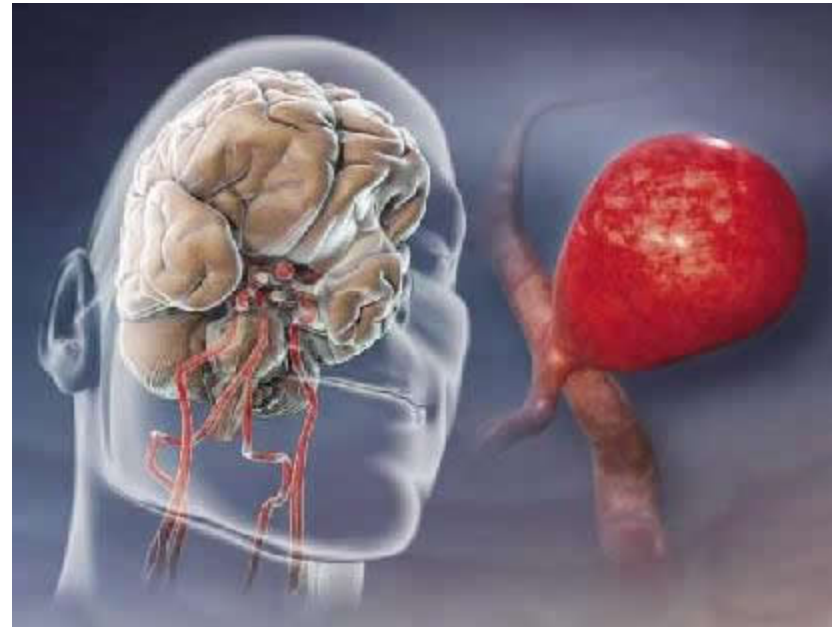


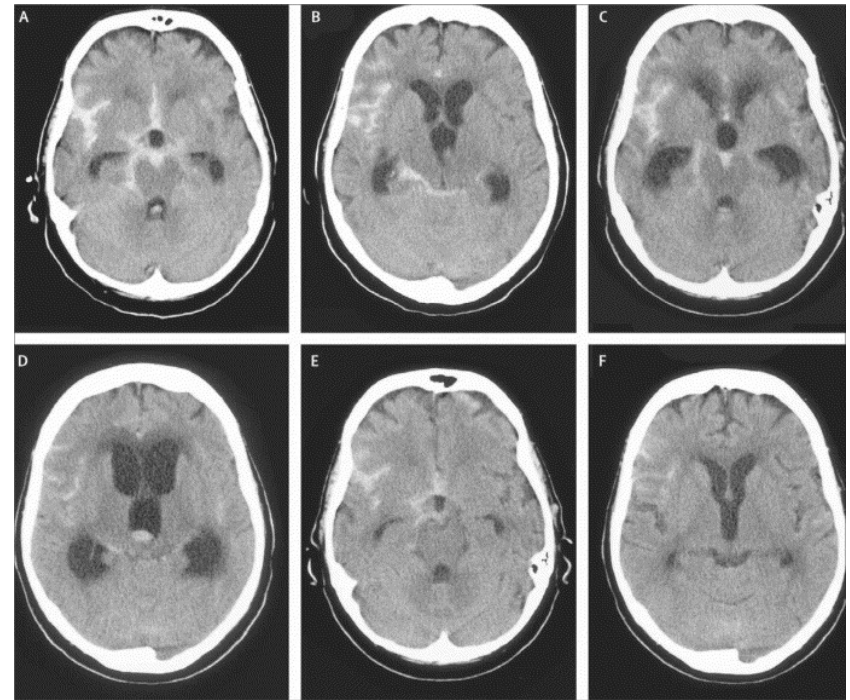
# **Subarachnoid hemorrhage**

**Dr. Ari Sami  
Neurosurgeon  
College of Medicine  
University of Sulaimani**



# SAH

**Bleeding from intracranial vessels lie in the SA space which give off small perforating branches to the brain tissue or from an associated aneurysm.**

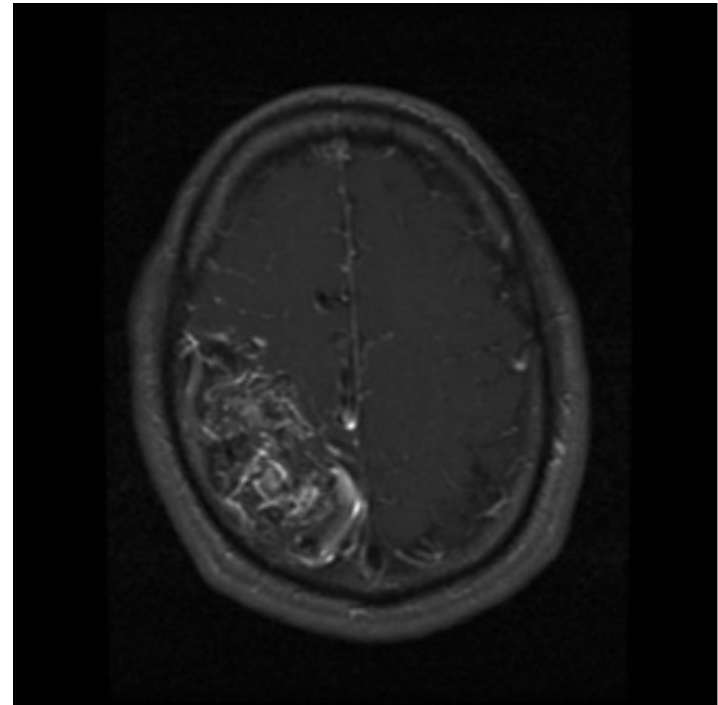
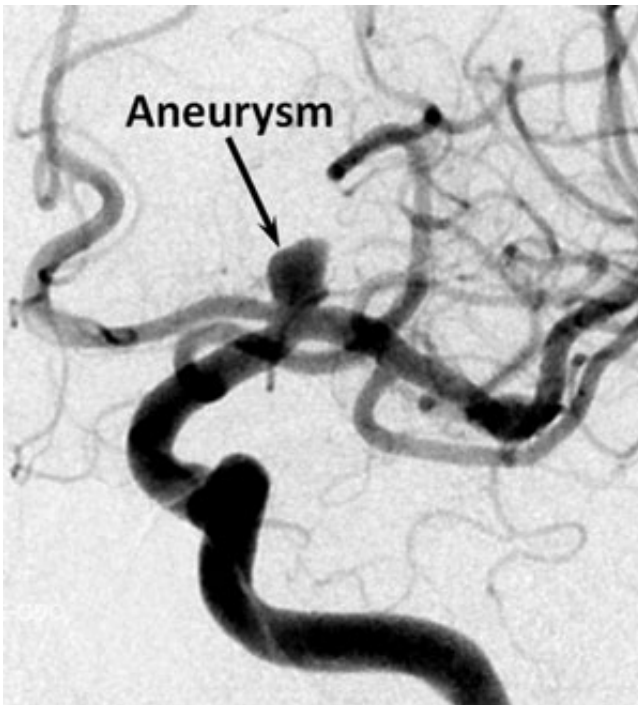


# Incidence

**It represents about 5-10% of all non-traumatic ICH with an incidence of approximately 10-15/100 000 population per year**

# **Causes of SAH**

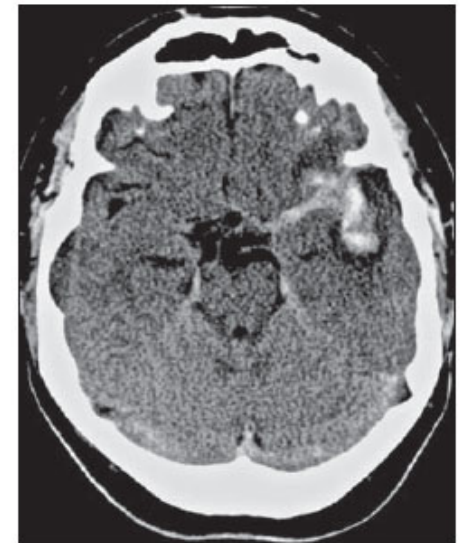
- Rupture of a berry aneurysm.
- Rupture of AVM (6%) (Children)



- **Rare Causes:**
  - **Bleeding from a tumor**
  - **Bleeding disorders**
  - **Blood dyscrasias**
  - **Rupture of spinal AVM**
  - **Undiagnosed cause in 15%**

# **Presenting features:**

- **Headache:**
  - Sudden onset of a severe headache of a type not experienced by the patient
  - Described as a blow to the head



- **Deteriorated level of consciousness**
- **Meningism**
  - Headache, fever, neck stiffness, photophobia, and vomiting



- **Focal neurological signs:**
  - **Due to:**
    - **Concomitant ICH:**
      - MCA (Temporal lobe and Parietal lobe), ACA (Frontal lobe)
    - **Local pressure effect of aneurysm (PCA and 3rd nerve palsy)**
    - **Cerebral vasospasm ( 2-3 days after the bleeding)**



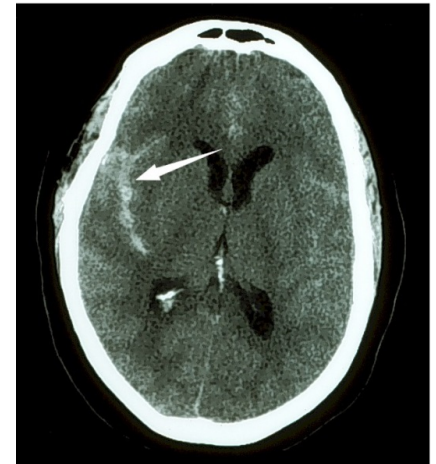
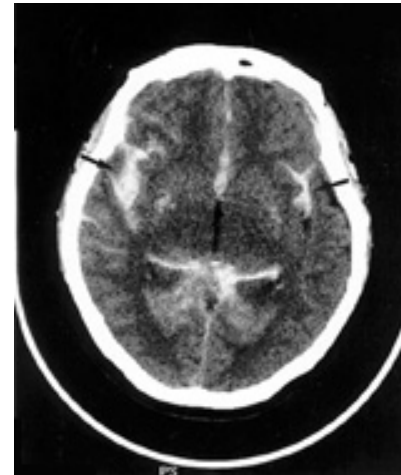
- **Epilepsy**
- **Optic fundi: papilloedema**
- **Retinal hemorrhage (subhyaloid)**
- **Reactive hypertension**
- **pyrexia**

# **Diagnosis:**

- **History and examination:**
  - Sudden severe headache
  - Decrease conscious state
  - Meningism

## **CT scan**

- It will show the hemorrhage in 95% of patients with SAH (if within 48 hours of the bleed)
- Hydrocephalus
- ICH
- Aneurysm or AVM
- Tumor

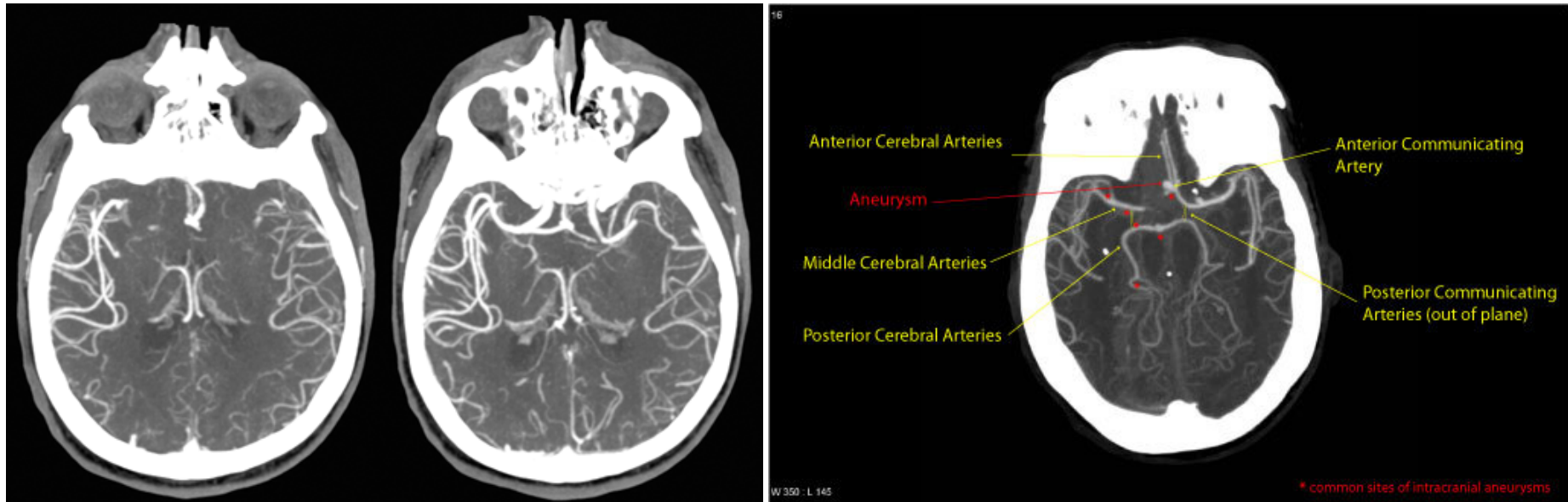


# **MRI**

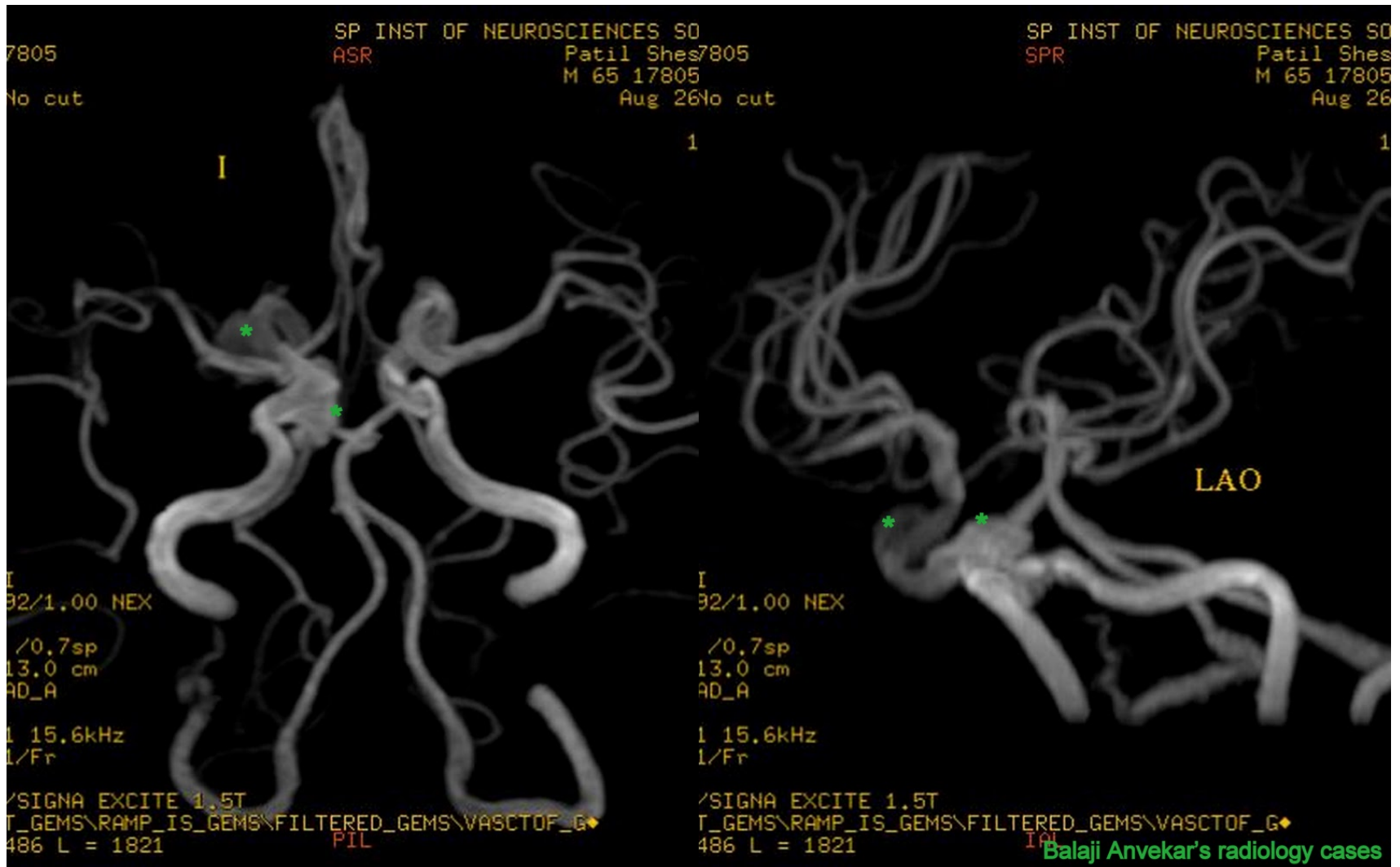
- **Not routinely used**
- **Used in spinal AVM which cause SAH**

# *CT angiography*

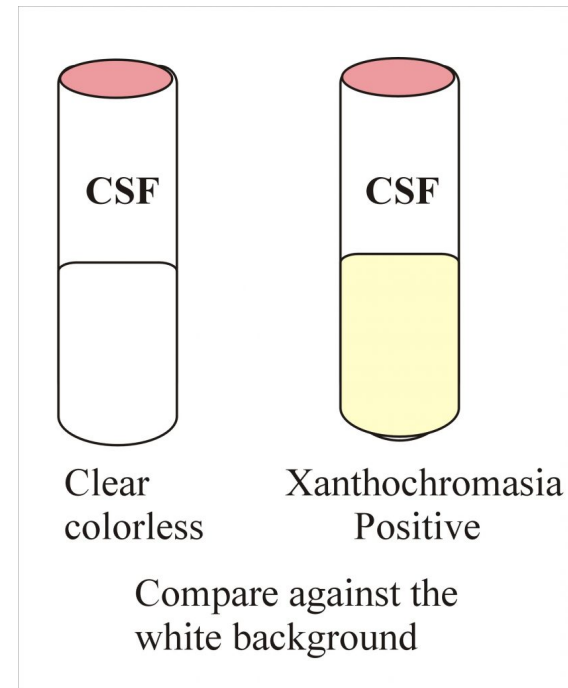
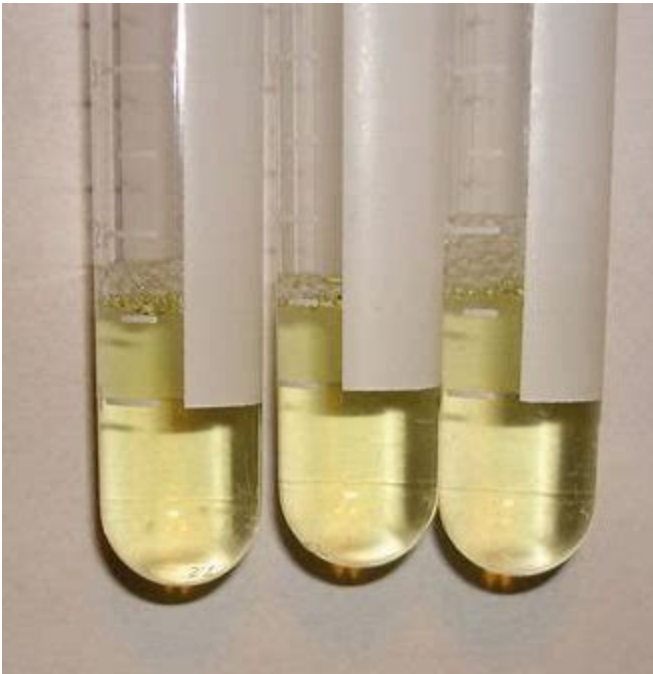
- Non-invasive techniques
- Will detect up to 95% of intracranial aneurysms



# MRA



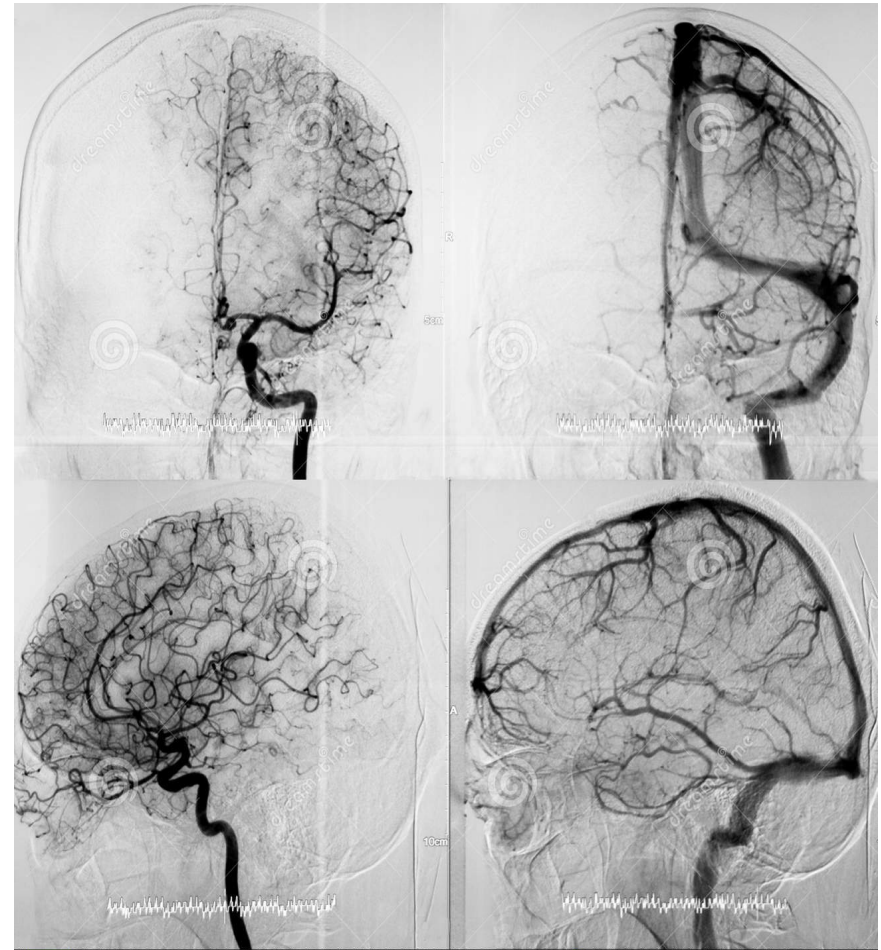
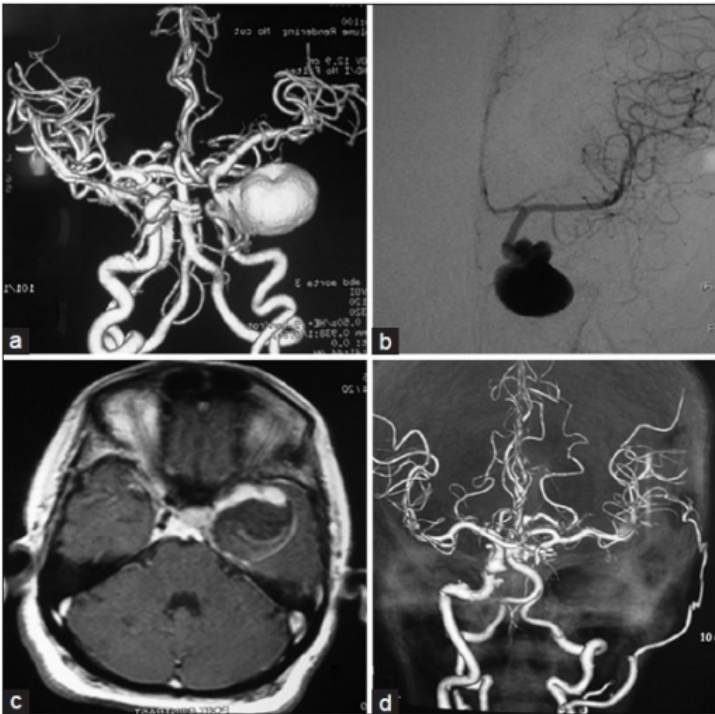
- **LP:**
  - **Xanthochromia:**
  - **Allow CSF to drip into 3 consecutive tubes**





# Cerebral angiography

- Aneurysm
- AVM
- Vessel spasm



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# **Aneurysm**

- **The most common cause of SAH with a maximum incidence in the 4th-5th decades of life**
- **The majority of aneurysms occur in constant positions of the circle of Willis and about 85% occur in the anterior half of the circle**

# **Clinical presentation**

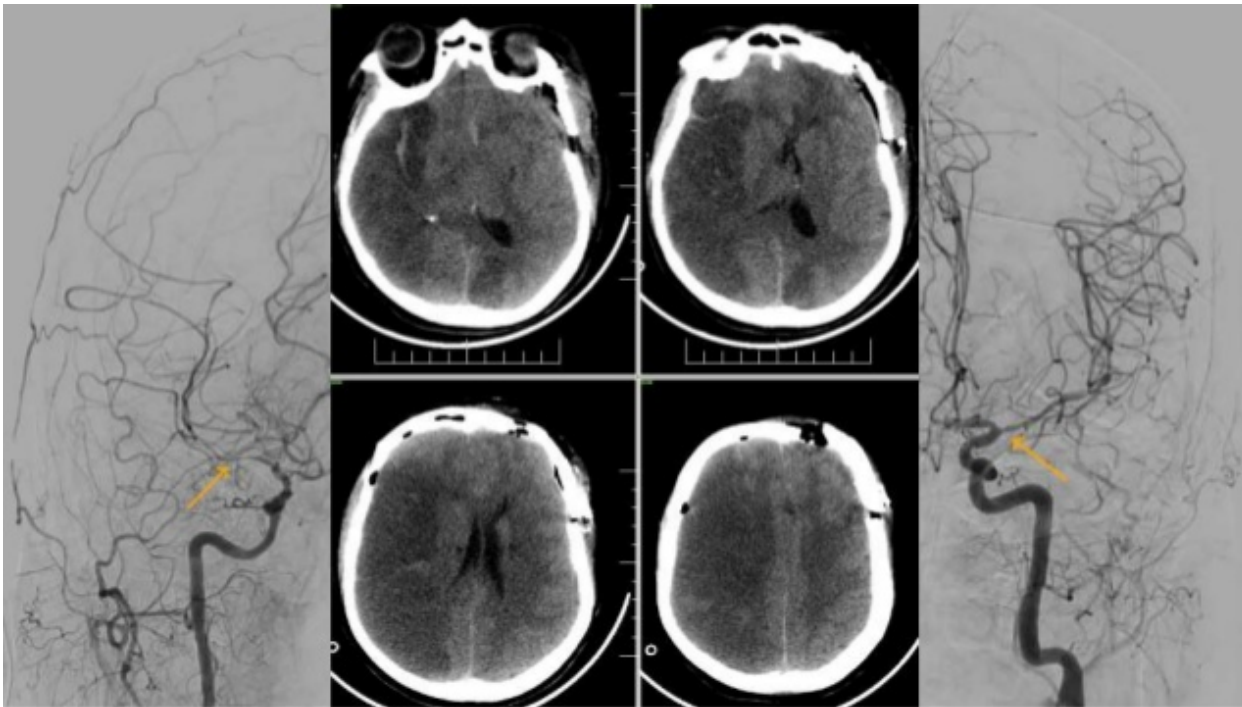
- **Rupture**
- **Compression from aneurysm sac**
  - Visual
  - Cranial nerves
  - Pituitary
  - Brainstem
- **incidental**



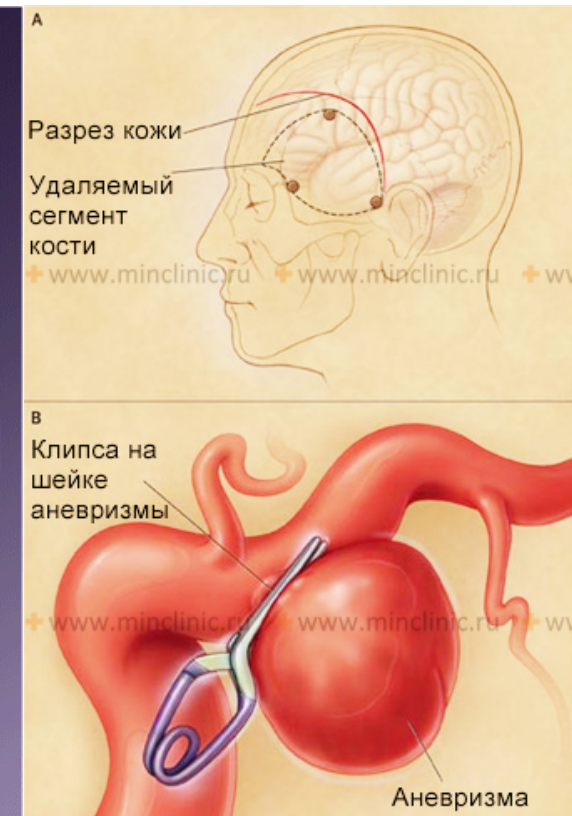
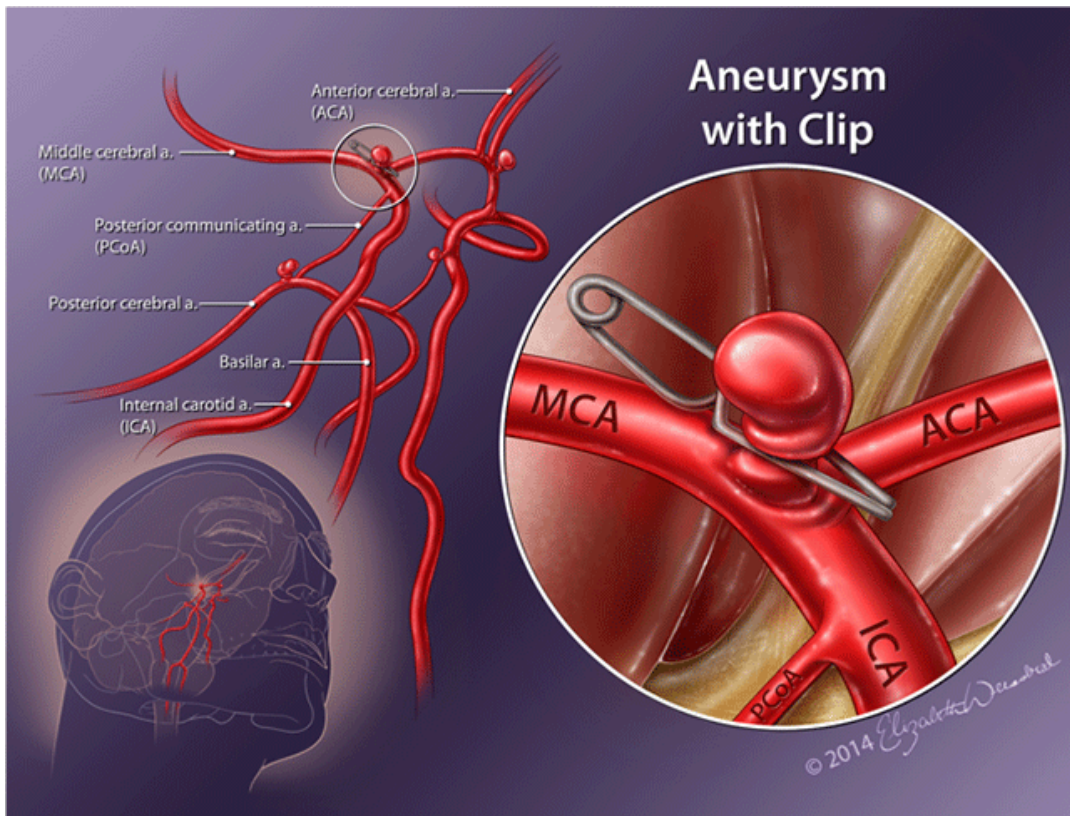
# **Management**

- **Severity of initial hemorrhage**
- **Rebleed: 50% within 6 weeks**
- **Antifibrinolytic agents are not used because of thrombotic phenomena (DVT, cerebral thrombosis)**

- **Cerebral vasospasm:**
  - Hypertensive and hypervolemic therapy
- **Calcium channel blocking agents (Nimodipine)**

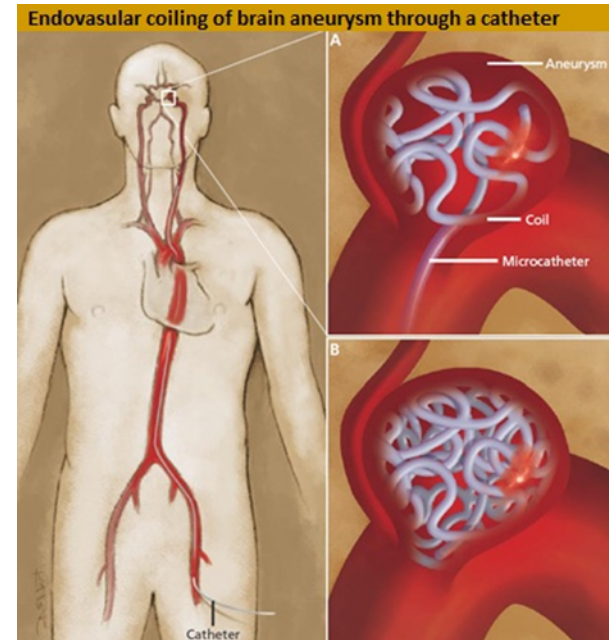


- **Surgery:**
  - **Occlusion of the neck**





- Reinforcement of the sac.
- Proximal ligation of the feeding vessel
- Endovascular procedures (detachable coils)



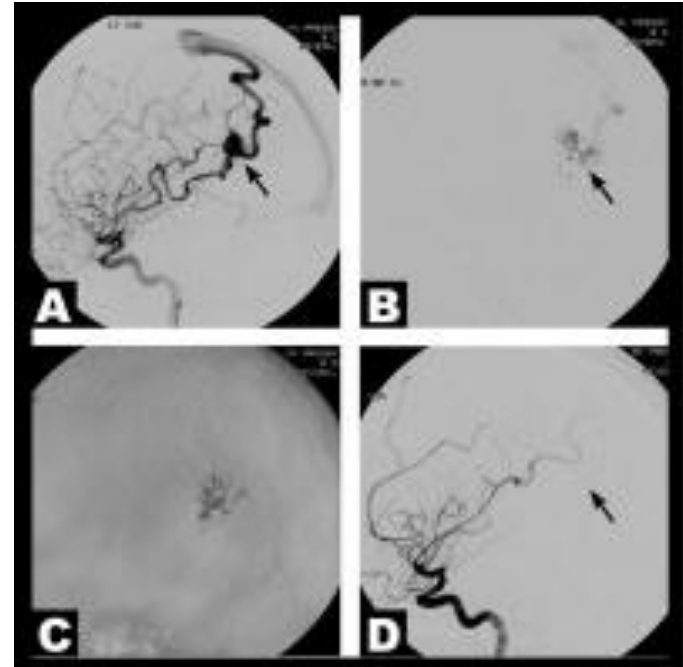
# **AVM**

- **Most common cause of SAH in children and young patients**
- **Clinical presentation:**
  - Hemorrhage
  - Epilepsy
  - Headache
  - Progressive neurological deficits



# **Management**

- **Excision**
- **Stereotactic radiosurgery**
- **Embolisation**







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Oct.2024

Dr.Ari S. Hussain Nadhim

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